

GOKUL P

Computer Science Engineering Graduate | Full Stack Development, AI Integration, Security & Enterprise Systems

Chennai, Tamil Nadu, India | gokul.p0807@gmail.com | +91 97914 52673

GitHub: github.com/lamgokul7 | Portfolio: gokulp-portfolio.vercel.app | LinkedIn: linkedin.com/in/gokulp0807

PROFESSIONAL SUMMARY

Computer Science Engineering graduate with hands-on experience across AI-powered application development, full-stack web development, secure authentication systems, and enterprise resource planning, built through six independent projects (including two AI-integrated platforms), four structured internships, and a peer-reviewed journal publication. Proficient in Python, FastAPI/Flask, JavaScript, REST API design, and relational database modeling, with practical exposure to LLM/AI API integration (Gemini), role-based access control, encryption, multi-tenant architecture, and PDF/document generation pipelines. Comfortable across the full development lifecycle — architecture, implementation, manual/regression testing, defect documentation, and deployment. Open to entry-level roles across Software Development, Full Stack/Web Development, AI/LLM Application Development, QA/Application Support, Data Analytics, Cybersecurity/SOC, SAP/ERP, and IT Analyst tracks, in both startups and established companies.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C++

Web & Backend: FastAPI, Flask, REST API design, HTML, CSS, PHP, Gunicorn, Jinja2, Starlette

AI/LLM Integration: Google Gemini API, google-genai SDK, prompt engineering, AI output validation/fabrication-checking, LLMpowered natural-language querying

Databases & Data: MySQL, SQLite — relational schema design, SQL (joins, aggregations, views), query optimization, data integrity validation, Pandas, NumPy, Excel

Document/PDF Engineering: WeasyPrint, xhtml2pdf, ReportLab, pypdf, python-docx, ATS-safe PDF formatting

Enterprise & Architecture: Multi-tenant system design, role-based access control (RBAC), SAP/ERP fundamentals, business process modeling

Security: AES encryption, DWT steganography, multi-factor authentication design, secure file workflows, access-control auditing, session/cookie security, IP-based rate limiting, Wireshark (network traffic analysis basics)

Cloud & DevOps: Oracle Cloud Infrastructure (OCI) — Certified Foundations Associate, Docker, Git/GitHub, Uvicorn

Testing & QA: Manual functional testing, regression testing, defect identification and reporting, SDLC practices

Tools & Design: XAMPP, Figma, OpenCV, Adobe Photoshop, Adobe Premiere Pro

PROJECTS

1. ResumeTailor AI — AI-Powered, ATS-Safe Resume Customization Engine

Jun 2026 – Jul 2026

Python, FastAPI, Google Gemini API, WeasyPrint, Jinja2 | GitHub: github.com/lamgokul7/ResumeTailor-AI | Live: resume-tailor-j5zl.onrender.com/login

- Architected and built a full-stack, closed-loop AI resume tailoring engine enforcing strict traceability, ensuring the LLM can only rephrase/select from a candidate's verified master content bank and never fabricate skills, credentials, or experience.
- Designed an end-to-end pipeline: resume/JD upload → text extraction (pypdf/python-docx) → Gemini-based keyword and JDmatch analysis → user-driven keyword selection → AI-customized rewrite of summary, skills, projects, and internships.
- Built an automated fabrication safety gate that tokenizes AI-generated output against a flattened master keyword bank, flagging any unrecognized/unverified technology terms as an amber warning badge before the resume is finalized.
- Implemented a resilient AI service layer with an automatic model failover chain to maintain uptime against rate limits and transient API errors.
- Engineered a paged PDF rendering pipeline using WeasyPrint with a Jinja2 templating layer (two-column and ATS-safe singlecolumn templates), including a graceful fallback to xhtml2pdf when GTK3 dependencies are unavailable.
- Built session-based authentication with signed cookies (Starlette/itsdangerous), custom middleware route guarding for all API endpoints, and IP-based rate limiting with lockout after repeated failed login attempts.
- Deployed the application live on Render, making it accessible as a fully functional hosted web app.

- Currently debugging and hardening the system: improving fabrication-check coverage across more violation types, updating a deprecated Gemini model reference in the JD match-scoring path, and fixing PDF hyperlink annotation rendering for full clickability.

2. ERPilot AI — Multi-Tenant Enterprise Resource Planning SaaS Platform

Mar 2026 – May 2026

Python, Flask, SQLite, Docker, Gemini API | github.com/lamgokul7/ERPilot-AI

- Architected and built a complete, AI-powered ERP SaaS platform end-to-end, covering employee management (HR), inventory tracking (Materials Management), supplier management, sales order processing (Sales & Distribution), notifications, and business reporting — directly analogous to core SAP HCM, MM, and SD module functions.
- Implemented multi-tenant architecture with tenant_id-based data isolation across a 7-table relational schema, securely supporting unlimited tenants on a single deployment, and systematically tested access boundaries across multiple tenant accounts to confirm correct data isolation.
- Built and audited role-based access control (RBAC) and authentication workflows from first principles, defining and verifying permission tiers across all 7 functional modules to prevent unauthorized cross-tenant data access.
- Designed and implemented REST-based backend modules in Flask for every core business function, manually testing API endpoints across iterations to identify and resolve access-control and data-integrity defects before each release.
- Wrote SQL queries using joins and aggregations to extract, filter, and validate data across modules, and built a business reporting module aggregating raw transactional data into decision-ready summary views.
- Integrated the Gemini API to power a context-aware AI assistant for natural-language business queries, with logging and review of every AI interaction to catch anomalous behavior.
- Containerized the full application with Docker and configured a production-ready deployment path using Gunicorn, considering deployment hardening and attack surface.

3. Enhanced Security Framework: Graphical Password Authentication with Data Hiding on Cloud Storage — Final Year Project (Published Research)

Jan 2025 – May 2025

Python, Flask, OpenCV, AES Encryption, DWT Steganography, SQLite | github.com/lamgokul7/Graphical-Password-Authentication-With-Data-Hiding

- Engineered a 3-factor authentication framework combining username/password validation, graphical image verification, and PIN-based authentication to strengthen resistance against credential-based attacks, monitoring login attempt patterns to distinguish valid from suspicious access.
- Implemented AES encryption for secure file protection and integrated DWT-based steganography for hidden data embedding, reducing exposure during file transfer.
- Designed and tested secure file upload, access-approval, and file-sharing workflows, validating correct rejection of unauthorized access attempts across invalid and boundary input cases.
- Built a full-stack web application integrating authentication modules, encrypted storage, NumPy-based pixel-level image processing, and database-driven user management.
- Maintained a structured, queryable log of authentication attempts and file-sharing events for traceability, verifying encryption/decryption integrity across repeated test cycles with zero data loss.
- Published as a peer-reviewed paper, "Enhanced Security Framework: Graphical Password Authentication with Data Hiding on Cloud Storage," in the International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE), Volume 14, Issue 4, April 2025. DOI: 10.17148/IJARCCE.2025.14481

4. Real Estate Management System — Team Project

Apr 2024 – May 2024

HTML, CSS, JavaScript, PHP, MySQL, XAMPP | github.com/lamgokul7/Real-Estate-Management-System

- Built a full property management platform (team of 3) supporting listings, search/filtering, bookings, and inquiry workflows, backed by a normalized MySQL schema with SQL joins for search, filtering, and reporting.
- Implemented user authentication and role-separated admin/customer modules, developing admin panels for managing properties, feedback, and transaction records.
- Designed responsive interfaces across devices and conducted interface validation, cross-device testing, and MySQL data consistency checks during team review cycles.

INTERNSHIP EXPERIENCE

Android App Development Intern — Ajashia Technologies

Feb 2025 – May 2025

- Built Android application features using Java and Kotlin, applying industry practices in Android development.
- Integrated Firebase backend services into application workflows, learning backend integration and modern app architecture.
- Took part in design, testing, and debugging cycles, developing problem-solving and analytical skills relevant to mobile development.

Frontend Web Development Intern — Motion Cut

Jan 2025 – Feb 2025

- Built responsive web interfaces in HTML/CSS, validated across desktop and mobile breakpoints.
- Designed UI layouts in Figma, then implemented them as functioning frontend components.
- Recognized for exceptional dedication, technical aptitude, and outstanding results.

Skill-Up Program Intern — Mitsogo Technologies

Jul 2024 – Aug 2024

- Contributed to wireframing, prototyping, and frontend implementation using HTML, CSS, and JavaScript within a collaborative team workflow.
- Completed the program under direct mentorship (guidance of Sanjay Kumar), receiving certification for consistent contribution and engagement throughout the internship.

Web Design Intern — Codebind Technologies

Jul 2023 – Aug 2023

- Designed and built responsive website layouts, focusing on cross-device usability and accessibility.
- Recognized with a certificate of outstanding completion for the internship program.

EDUCATION

Bachelor of Engineering, Computer Science and Engineering

Nov 2021 – May 2025

Anand Institute of Higher Technology, Anna University | CGPA: 7.84 / 10.00

Higher Secondary Certificate (XII) — A.V.Rm.V Matric Higher Secondary School

Jun 2020 – May 2021

Score: 81.93%

Secondary School Certificate (X) — A.V.Rm.V Matric Higher Secondary School

Jun 2018 – May 2019

Score: 71.6%

PUBLICATIONS

- Enhanced Security Framework: Graphical Password Authentication with Data Hiding on Cloud Storage — Published in IJARCCCE (International Journal of Advanced Research in Computer and Communication Engineering), Volume 14, Issue 4, April 2025. DOI:10.17148/IJARCCCE.2025.14481

CERTIFICATIONS

- Oracle Cloud Infrastructure 2023 Certified Foundations Associate — Oracle (Apr 2024)
- Journey to Cloud: Envisioning Your Solution — IBM SkillsBuild (Sep 2024)
- MongoDB Node.js Developer Path for SmartBridge — MongoDB (Sep 2024)
- Networking Essentials — Cisco Networking Academy (May 2023)
- Learn Figma: UI/UX Design Masterclass — Udemy (Nov 2024)